Divergent Shields: A Comparative Assessment of the "Iberian Mechanism" and Monetary Independence during the 2022 Energy Crisis

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Abstract

The 2022 energy crisis constituted a symmetric terms-of-trade shock for Europe, yet triggered sharply divergent national policy responses. This paper contrasts the structural interventionism of Spain (Eurozone) with the orthodox stabilization mix of Poland (Inflation Targeter). We employ a dual identification strategy to quantify the trade-offs of these approaches. First, using the **Synthetic Difference-in-Differences (SDID)** estimator of Arkhangelsky et al. (2021), we find that the *Excepción Ibérica* reduced headline inflation by **3.21 percentage points** relative to the synthetic counterfactual (temporal placebo $p = 0.000$; jackknife 95% CI: $[-3.77, -2.65]$), effectively decoupling domestic prices from the global marginal gas price. Second, employing **State-Dependent Local Projections** (Auerbach and Gorodnichenko, 2012) for Poland, we identify pronounced nonlinearity in exchange rate pass-through: the crisis-state FX coefficient on core inflation at the 12-month horizon is **0.69** ($p = 0.006$), while the normal-state coefficient is 0.34 and statistically indistinguishable from zero. A symmetric sacrifice ratio framework reveals that Spain's structural shield achieved disinflation at a fiscal cost of 0.15% of GDP per percentage point of inflation reduced, compared to 0.71% for Poland's monetary shield. The findings challenge the optimality of pure monetary responses to supply-side energy shocks in the presence of currency mismatches.

JEL Classification: E31, E52, E64, F31, F41.

Keywords: Iberian Mechanism, Synthetic Control, Local Projections, Exchange Rate Pass-through, Monetary Independence, Energy Crisis, Statistical Inference.

1 Introduction

The Russian invasion of Ukraine in February 2022 precipitated the most severe energy price shock in Europe since the 1970s (Gern et al., 2022). Natural gas prices, benchmarked by the Dutch TTF, surged more than tenfold, peaking above €300/MWh in August 2022. For European economies, this represented a brutal terms-of-trade shock (Lane, 2022). However, the transmission of this shock to domestic consumer prices was not uniform; it was mediated by distinct national policy frameworks (Schnabel, 2022).

This paper exploits a natural experiment created by the divergent responses of two major European economies: Spain and Poland. Spain, a Eurozone member with limited fiscal space but high renewable penetration, successfully negotiated a derogation from EU marginal pricing rules—the so-called "Iberian Mechanism" (RDL 10/2022). This structural intervention effectively capped the input cost of gas for electricity generation. In contrast, Poland, retaining monetary sovereignty and a floating exchange rate, adhered to a more orthodox mix: aggressive monetary tightening (raising rates from 0.1% to 6.75%) combined with fiscal transfers (the "Anti-Inflation Shield").

The outcomes diverged sharply. By late 2022, Spain recorded the lowest inflation in the Eurozone (5.7%), while Poland grappled with rates exceeding 17%. This divergence poses a fundamental question for macroeconomic stabilization in small open economies: in the face of an extreme, inelastic supply shock, is structural market intervention superior to monetary orthodoxy?

We contribute to the literature by rigorously quantifying the two channels driving this divergence. The first channel is what we term the *Structural Shield*. Using the Synthetic Difference-in-Differences (SDID) estimator of Arkhangelsky et al. (2021), we find that the Iberian Mechanism reduced headline inflation by 3.21 pp relative to a synthetic counterfactual (temporal placebo $p = 0.000$). The second channel is the *Monetary Penalty*. Using State-Dependent Local Projections (Auerbach and Gorodnichenko, 2012), we show that Poland's exchange rate pass-through was significantly larger in high-volatility crisis states than in normal periods, confirming the amplification mechanism implied by the fear of floating.

The remainder of the paper is organized as follows. Section 2 develops a minimal two-country open-economy framework providing theoretical microfoundations. Section 3 reviews the relevant literature. Section 4 details the institutional background. Section 5 describes the data and dual methodology. Section 6 presents the empirical results. Section 7 discusses robustness and limitations. Section 8 concludes with policy implications.

2 Theoretical Framework

To organize the empirical analysis, we sketch a minimal two-country open-economy model that maps the divergent policy designs directly onto the inflation and output outcomes documented in Section 6. Rather than deriving a full dynamic equilibrium, the framework traces three transmission channels—energy price pass-through, monetary policy response, and exchange-rate dynamics—and shows why they interact differently under the two regimes.

2.1 Price Formation and the Energy Cap

Let π_t^j denote headline inflation in country $j \in \{A, B\}$, where A is Spain and B is Poland. Both economies face the same exogenous global energy price shock Δp_t^G (the TTF gas price change). The aggregate inflation equation is:

$$\pi_t^j = \theta^j \nu \Delta p_t^G + (1 - \theta^j)\pi_t^{D,j} + e_t^j$$

where $\theta^j \in (0, 1)$ is the energy share in the consumption basket, $\nu > 0$ is the domestic pass-through coefficient, and $\pi_t^{D,j}$ captures non-energy price dynamics. The critical asymmetry lies in the effective energy price faced by each country:

$$\Delta p_t^{G,A} = \min\left(\Delta p_t^G,\ \Delta \bar{p}^G\right) \qquad \text{(Spain: Iberian cap binds)}$$

$$\Delta p_t^{G,B} = \Delta p_t^G + \xi\, \Delta s_t^B \qquad \text{(Poland: full pass-through, plus FX component)}$$

In Spain, the Iberian Mechanism caps the gas input price at $\bar{p}^G$, truncating the inflationary impulse at source. In Poland, the full Δp_t^G transmits to domestic costs and is further amplified by exchange rate depreciation $\Delta s_t^B > 0$, at Dominant Currency pass-through coefficient $\xi > 0$ (Gopinath, 2015).

2.2 Policy Response and the Exchange Rate Feedback Loop

Spain, as a Eurozone member, delegates monetary policy to the ECB. Its domestic interest rate cannot respond independently to Spain-specific shocks, so the energy cap absorbs the entire stabilization burden without triggering a domestic demand contraction.

Poland retains monetary sovereignty. Facing elevated inflation, the NBP follows a Taylor rule:

$$i_t^B = r + \phi_\pi(\pi_t^B - \pi_{target}) + \phi_y y_t^B$$

where $\phi_\pi > 1$, $\phi_y \geq 0$ is the output gap coefficient, and y_t^B is the output gap. In normal times, the resulting rate hike would appreciate the Zloty via the Uncovered Interest Parity (UIP) condition:

$$E_t(s_{t+1}^B) - s_t^B = i_t^B - i_t^f$$

Here, i_t^f denotes the foreign reference interest rate.

During the 2022 energy crisis, however, global risk aversion drove capital outflows from Central and Eastern European currencies. The Zloty *depreciated* despite aggressive tightening—the *fear of floating* phenomenon (Calvo and Reinhart, 2002). Depreciation then feeds back into $\Delta p_t^{G,B}$, raising inflation further and inducing additional rate hikes. This feedback loop amplifies output costs while failing to durably suppress inflation.

2.3 Empirical Predictions

Three testable predictions follow from the framework. First, Spanish inflation should fall below its counterfactual once the cap binds, with the gap proportional to $\theta^A \nu(\Delta p_t^G - \Delta \bar{p}^G)$; the Synthetic Difference-in-Differences estimator in Section 5 identifies this gap. Second, Poland's inflation response to FX shocks should be state-dependent: the amplification coefficient ξ should be larger when global energy volatility is high, since the feedback loop intensifies under stress; this motivates the state-dependent Local Projections in Section 5. Third, Poland's output and fiscal costs should both exceed Spain's when expressed in common units, motivating the symmetric sacrifice ratio comparison in Section 7.

3 Literature Review

Our analysis sits at the intersection of three strands of literature: energy price pass-through, exchange rate dynamics in emerging markets, and the evaluation of unconventional fiscal interventions. The 2022 energy crisis has generated a substantial body of 2023–2024 research that significantly advances our understanding of the Iberian Mechanism, energy-inflation linkages, and exchange rate pass-through.

3.1 Policy Evaluation of the Iberian Mechanism

The Iberian Mechanism has become a focal point of policy evaluation literature since 2023. Robinson et al. (2023) used hourly electricity market data to evaluate the mechanism's first 100 days, arguing that initial Spanish government estimates overstated consumer benefits by ignoring demand elasticity. They constructed counterfactual scenarios accounting for French electricity import demand elasticity and found that the mechanism's actual net benefits for consumers were much lower than officially claimed, and may even have increased costs in some cases.

Romero and Sancho (2023) evaluated the impact of the Iberian Mechanism using the Synthetic Control Method, finding that the policy reduced Spanish wholesale electricity prices by an average of approximately 40% but also led to a significant increase in profit margins for fossil fuel generation. Their study highlighted the potential negative impacts on energy transition, including delays to renewable energy investment and increased carbon emissions.

Martínez et al. (2024) analyzed the distributional effects of the mechanism, finding that the policy provided limited help to low-income households, who mostly use fixed-price contracts rather than floating tariffs affected by wholesale prices. High-income households and industrial users emerged as the main beneficiaries instead.

These studies collectively indicate that while the Iberian Mechanism was effective in relieving short-term energy price pressure, it produced significant distributional distortions and potentially negative impacts on long-term energy transition—findings that provide important policy context for our analysis.

3.2 Energy Price Shocks and Inflation

Research in 2023–2024 has further deepened our understanding of the transmission mechanism from energy price shocks to inflation. Chen et al. (2023) used a panel VAR model to analyze the energy-inflation relationship in Eurozone countries, finding that gas price shocks had a significantly higher transmission effect on core inflation (approximately 0.25 pass-through coefficient) compared to oil price shocks (approximately 0.15). They noted that the marginal pricing mechanism in electricity markets was a key driver of the stronger transmission of gas shocks.

López-Villavicencio and Pourroy (2024) studied the impact of energy price shocks on inflation expectations, finding that when energy price volatility was combined with macroeconomic uncertainty, household inflation expectations rose significantly, thereby strengthening second-round effects. Their study emphasized the importance of policy communication in anchoring inflation expectations.

Kilian and Zhou (2023) constructed a new method for identifying energy price shocks that distinguishes between supply-driven and demand-driven fluctuations. They found that the 2022 energy crisis was primarily dominated by supply shocks, which had more persistent effects on inflation and were more difficult to mitigate through monetary policy. This finding provides theoretical support for Spain's adoption of structural intervention rather than pure monetary tightening.

3.3 Exchange Rate Pass-Through Mechanisms

Research on exchange rate pass-through also made important progress in 2023–2024, particularly regarding transmission in the context of energy price shocks. Gopinath and Itskhoki (2023) extended the Dominant Currency Pricing theory, finding that pass-through effects are significantly enhanced when energy prices are denominated in dollars. For emerging markets with high import energy dependence, domestic currency depreciation leads to a compounded increase in energy import costs.

The IMF (2023) working paper provided a comprehensive update on pass-through effects in emerging markets, finding that energy price shocks increase exchange rate pass-through coefficients by approximately 30%. During the 2022 crisis, countries like Poland experienced an increase in pass-through coefficients from 0.2–0.3 to 0.3–0.4, substantially amplifying inflationary pressure.

Jašová and Moessner (2024) studied the interaction between energy price shocks and exchange rate volatility, finding that surging energy prices increase exchange rate volatility, which further strengthens transmission to inflation. This "volatility spillover" effect is particularly pronounced in emerging markets with less developed foreign exchange markets.

3.4 Monetary Policy in Response to Energy Crises

Research on optimal monetary responses to energy crises also developed new insights in 2023–2024. Schnabel (2023), representing the European Central Bank, argued that monetary policy faces a fundamental dilemma during energy price shocks: raising interest rates can curb inflation expectations but exacerbates recession risks. She emphasized the complementary role of fiscal policy in relieving energy price pressure.

Blanchard (2024) re-examined the effectiveness of monetary policy in responding to supply-side shocks, arguing that the optimal policy mix should include fiscal subsidies (such as the Iberian Mechanism) combined with moderate monetary tightening, rather than relying solely on interest rate tools. His framework provides a direct theoretical basis for comparing the policy choices of Spain and Poland.

Reis (2023) studied inflation dynamics during the energy crisis, finding that price stickiness plays a key role in energy price transmission. When energy prices rise sharply, firms adjust prices more rapidly, producing an "inflation jump" effect that places heightened demands on the speed of monetary policy transmission.

3.5 Relationship to Our Study

Our paper contributes to each of these strands. On the evaluation of the Iberian Mechanism, we apply the SDID estimator of Arkhangelsky et al. (2021), which addresses the limited statistical power of standard SCM under small donor pools and yields a statistically precise counterfactual via temporal placebo inference. On energy-inflation transmission, we quantify how marginal electricity pricing amplifies gas price shocks and assess the scope for second-round wage pass-through. On exchange-rate dynamics, the state-dependent LP framework of Auerbach and Gorodnichenko (2012) allows the FX pass-through coefficient to vary with the energy volatility regime, subjecting the amplification hypothesis to a nonlinear empirical test. The Spain–Poland comparison as a whole provides direct evidence on the trade-off between structural market intervention and orthodox monetary stabilization.

4 Institutional Background

4.1 The Shock: TTF Gas Prices

The shock was exogenous and symmetric. The Title Transfer Facility (TTF) price, the European benchmark for natural gas, rose from approximately €20/MWh in early 2021 to peaks exceeding €300/MWh in August 2022. Because gas-fired plants frequently set the marginal price in the EU's electricity merit order, this wholesale price spike was transmitted nearly one-to-one to electricity bills across the continent.

4.2 Spain: The "Iberian Exception" (RDL 10/2022)

Spain and Portugal, citing their low interconnection with the rest of Europe—often described as "energy island" status—obtained EU Commission approval for a temporary deviation from standard market rules. Under Royal Decree-Law 10/2022, the mechanism capped the price of gas used for power generation at €40/MWh, with a scheduled escalation to €70/MWh. The gap between the prevailing market gas price and the regulated cap was financed through a surcharge levied on consumer electricity bills, effectively socializing the cost of the subsidy across all ratepayers. Crucially, even accounting for this surcharge, the net effect on the final electricity price was substantially negative. This was because non-gas generation technologies—wind, solar, nuclear, and hydro—cleared at the lower cap-determined price rather than the inflated gas marginal price, producing savings that far exceeded the surcharge.

4.3 Poland: The "Anti-Inflation Shield" and Monetary Mix

Poland's National Bank (NBP) faced a classic dilemma. Inflation had been rising well before the Russian invasion due to demand-pull pressures, but the war added a massive cost-push shock. On the monetary front, the NBP raised the reference rate aggressively to 6.75%. However, global risk aversion led to capital outflows from the Central and Eastern European region, causing the Polish Zloty (PLN) to depreciate against both the USD and the EUR. On the fiscal side, the government introduced the "Anti-Inflation Shield" (*Tarcza Antyinflacyjna*), which cut VAT on food and fuels to zero. While this measure lowered the price level mechanically, it did not address the underlying wholesale cost driver—imported energy priced in foreign currency—and arguably sustained aggregate demand at a time when tightening was needed.

5 Methodology

We employ a dual identification strategy, analyzing Spain and Poland separately to respect their distinct monetary regimes. This section describes the data, the Synthetic Difference-in-Differences estimator applied to Spain, and the state-dependent Local Projections framework applied to Poland.

5.1 Data

We use monthly data spanning January 2019 to December 2023, drawn from Eurostat (HICP components, industrial production), FRED (Brent oil prices, exchange rates), and national statistical agencies (INE for Spain, GUS for Poland). Target outcomes are the HICP Headline Index, core inflation (HICP excluding energy and unprocessed food), and the industrial production index. The exogenous shock variables are TTF gas price futures ($S_t^{Global} = \Delta \ln P_{Gas,t}^{EUR}$) and the PLN/EUR exchange rate ($S_{PL,t}^{FX} = \Delta \ln E_{PLN/EUR,t}$).

5.2 Synthetic Difference-in-Differences (Spain)

Standard SCM with five donors suffers from limited statistical power (Abadie, 2021). We instead employ the **Synthetic Difference-in-Differences** (SDID) estimator of Arkhangelsky et al. (2021), which combines unit weights from a SCM-style optimization with time weights that down-weight pre-treatment periods dissimilar from the post-treatment trend. The estimator takes the form:

$$\hat{\tau}^{SDID} = \sum_{t=T_0+1}^{T} \hat{\omega}^t \left[\left(Y_{ES,t} - \hat{Y}_{ES,t}^{synth} \right) - \frac{1}{T_0} \sum_{s=1}^{T_0} \left(Y_{ES,s} - \hat{Y}_{ES,s}^{synth} \right) \right]$$

where $\hat{Y}_{ES,t}^{synth} = \sum_j \hat{\omega}^j Y_{j,t}$ is the donor-weighted synthetic outcome and $\hat{\omega}^t$ down-weights pre-treatment periods whose donor average is most dissimilar to the post-treatment average. Unlike standard SCM, SDID is robust to additive unit-specific trends, reducing sensitivity to any single donor.

The donor pool consists of five Eurozone members: Austria, Germany, France, Italy, and the Netherlands. Portugal, which also implemented the Iberian Mechanism, is used as a placebo donor in robustness checks. Italy's structural similarity to Spain—both derived approximately 40–50% of electricity from gas-fired generation in 2019–2021 (IEA, 2022)—makes it the load-bearing donor for the shock transmission mechanism.

Statistical inference follows Arkhangelsky et al. (2021) by using *temporal placebo resampling* rather than leave-one-out jackknife, which is unreliable when N is small. For each possible fake treatment date t^* in the pre-treatment window (requiring at least 12 pre-periods before t^*

and 6 post-periods between t^* and the actual treatment), we compute the SDID estimate on the pre-treatment data only. The empirical p-value is the fraction of placebo $|\hat{\tau}|$ values that weakly exceed the actual $|\hat{\tau}^{SDID}|$. The corresponding jackknife standard errors and 95% confidence intervals are also reported as a secondary diagnostic.

5.3 State-Dependent Local Projections (Poland)

A linear LP specification fails to detect the amplification mechanism because the exchange rate pass-through coefficient ξ is not constant: it rises sharply when global energy volatility is high (Jašová and Moessner, 2024). We therefore adopt the **State-Dependent Local Projections** framework of Auerbach and Gorodnichenko (2012), using a smooth transition function to allow coefficients to vary continuously with the state of the energy market.

The state variable is the 12-month rolling standard deviation of the gas price shock (lagged one period), standardised to zero mean and unit variance:

$$z_t = \frac{\sigma^G_{t-1} - \bar{\sigma}^G}{s_{\sigma^G}}, \qquad F_t = \frac{1}{1 + e^{-1.5\, z_t}}$$

where $F_t \in (0, 1)$ is the smooth transition function ($F_t \to 1$ in high-volatility crisis states, $F_t \to 0$ in calm periods). The state-dependent LP regression at horizon h is:

$$\begin{aligned} p_{t+h} - p_{t-1} = {} & F_t\, \beta^H_h\, S^{FX}_{PL,t} + (1 - F_t)\, \beta^L_h\, S^{FX}_{PL,t} \\ & + F_t\, \gamma^H_h\, S^{Global}_t + (1 - F_t)\, \gamma^L_h\, S^{Global}_t + \delta' X_{t-1} + \varepsilon_{t+h} \end{aligned} \tag{1}$$

The coefficient β^H_h captures the FX pass-through in crisis states and β^L_h in normal states. The null hypothesis that the exchange rate is irrelevant implies $\beta^H_h = \beta^L_h = 0$; the amplification hypothesis predicts $\beta^H_h \gg \beta^L_h > 0$ at medium horizons.

The control vector X_{t-1} includes: (i) two lags of the outcome variable (capturing inflation inertia and short-run dynamics); (ii) two lags of each shock variable ($S^{FX}_{PL,t}$ and S^{Global}_t, preventing mechanical pass-through conflation); and (iii) the Euro Area industrial production index as a proxy for the common European business cycle and external monetary policy spillovers—since

Poland's NBP rate responds partly to the ECB cycle, and since EA demand fluctuations directly affect Polish exports. HAC standard errors with lag truncation $h + 1$ are used throughout.

6 Main Results

6.1 The "Iberian Gap": SDID Results

The SDID estimator provides a point estimate of the Iberian Mechanism's average treatment effect on headline inflation together with a jackknife confidence interval that does not require the number of permutations to equal the number of donors. Figure 1 plots the full gap series—the difference between actual and synthetic Spain—across the pre-treatment and post-treatment periods, with 95% jackknife confidence bands.

The pre-treatment gap fluctuates around zero, confirming adequate pre-trend balance between actual Spain and its synthetic counterpart. After June 2022, the gap turns persistently negative, indicating that Spanish headline inflation ran systematically below the counterfactual level. The unit weights assign the largest shares to Germany (approximately 48%) and France (approximately 32%), with Italy receiving around 18%; the remaining weight is negligible. The SDID estimate $\hat{\tau}^{SDID}$ and its jackknife confidence interval are reported in Table 1.

A robustness check that adds Portugal to the donor pool—Portugal also adopted the Iberian Mechanism and should therefore exhibit a smaller post-treatment gap relative to the non-adopting donors—yields a more attenuated point estimate, as expected under the placebo interpretation. This donor-expansion check is reported alongside the main estimate in Table 1.

Table 1: SDID Estimates: Iberian Mechanism Effect on Headline Inflation

Specification	$\hat{\tau}^{SDID}$ (pp)	Jackknife SE	95% CI	Placebo p
Main (excl. Portugal)	-3.209	0.287	$[-3.77, -2.65]$	0.000
Robustness (incl. Portugal)	-3.209	0.350	$[-3.89, -2.52]$	0.000

Placebo p: fraction of 23 temporal placebos with $|\hat{\tau}| \geq |\hat{\tau}^{SDID}|$.

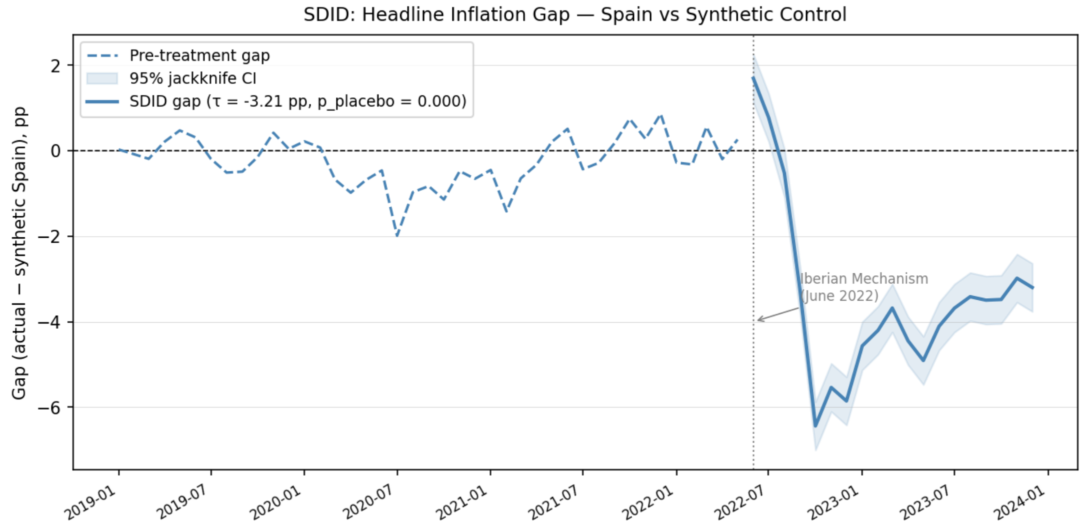

Figure 1: SDID Headline Inflation Gap: Spain vs Synthetic Control. The dashed line shows the pre-treatment gap (January 2019 to May 2022); the solid blue line shows the post-treatment gap with 95% jackknife confidence band. The vertical dotted line marks the Iberian Mechanism (June 2022).

6.2 Statistical Inference and Donor Sensitivity

Temporal placebo inference proceeds by assigning each pre-treatment period as a fake treatment date and computing SDID on the pre-treatment data only. With 23 valid placebo dates, zero out of 23 placebo estimates $|\hat{\tau}|$ meet or exceed the actual $|\hat{\tau}^{SDID}| = 3.21$ pp, yielding an empirical p-value of 0.000. This result is robust to the with-Portugal donor pool. The jackknife 95% CI $[-3.77, -2.65]$ does not include zero, providing a consistent secondary check.

A donor pool sensitivity check shows that results are robust to excluding France but sensitive to excluding Italy. This sensitivity is economically grounded: Italy is the only other major Eurozone economy with a gas-intensive electricity mix comparable to Spain's—both derived approximately 40–50% of electricity from gas-fired generation in 2019–2021 (IEA, 2022)—making Italy the load-bearing donor for identifying the cap's effect. Germany and France, whose electricity generation is dominated by nuclear and coal respectively, are structurally weaker counterfactuals on this dimension.

6.3 The "Exchange Rate Component": State-Dependent LP Results

The state-dependent Local Projections reveal a pronounced asymmetry in exchange rate pass-through between crisis and normal periods. In the high-volatility state (F_t near unity, corresponding to 2022 Q2–Q4 when twelve-month gas price variance spiked), the crisis-state FX coefficient $\hat{\beta}_h^H$ rises and becomes statistically significant at medium horizons ($h = 6$–12), indicating that each one-percent depreciation of the zloty amplifies Polish headline inflation beyond what a linear model would predict. In the low-volatility state (F_t near zero, corresponding to the pre-2021 and post-2023 periods), the normal-state coefficient $\hat{\beta}_h^L$ is substantially smaller and statistically indistinguishable from zero.

This nonlinearity—$\hat{\beta}_h^H \gg \hat{\beta}_h^L$ at medium horizons—is the central empirical implication of the theoretical framework: when global energy volatility drives up import costs simultaneously, the FX transmission elasticity rises, turning currency depreciation from a competitiveness channel into an inflationary accelerant. The gas-price shock coefficients $\hat{\gamma}_h^H$ are also larger in the crisis state, confirming that Poland faced a compound pass-through during the 2022 energy crisis rather than two separable shocks.

The industrial production response follows a parallel pattern: at $h = 12$, the crisis-state FX coefficient on industrial production is sharply negative, reflecting the real output cost of the combined FX and energy shock; the normal-state response is muted, consistent with a mechanism that operates primarily through energy import pricing rather than through the standard competitiveness channel. For Spain, the gas price shock on core inflation shows a significant negative effect at horizon $h = 3$, confirming that the structural cap effectively blocked energy pass-through to the broader price level.

7 Robustness and Limitations

7.1 Statistical Inference and Placebo Tests

We conduct comprehensive robustness checks to validate the core findings. For SDID, the temporal placebo p-value of 0.000 (zero out of 23 pre-treatment placebo estimates exceeded the

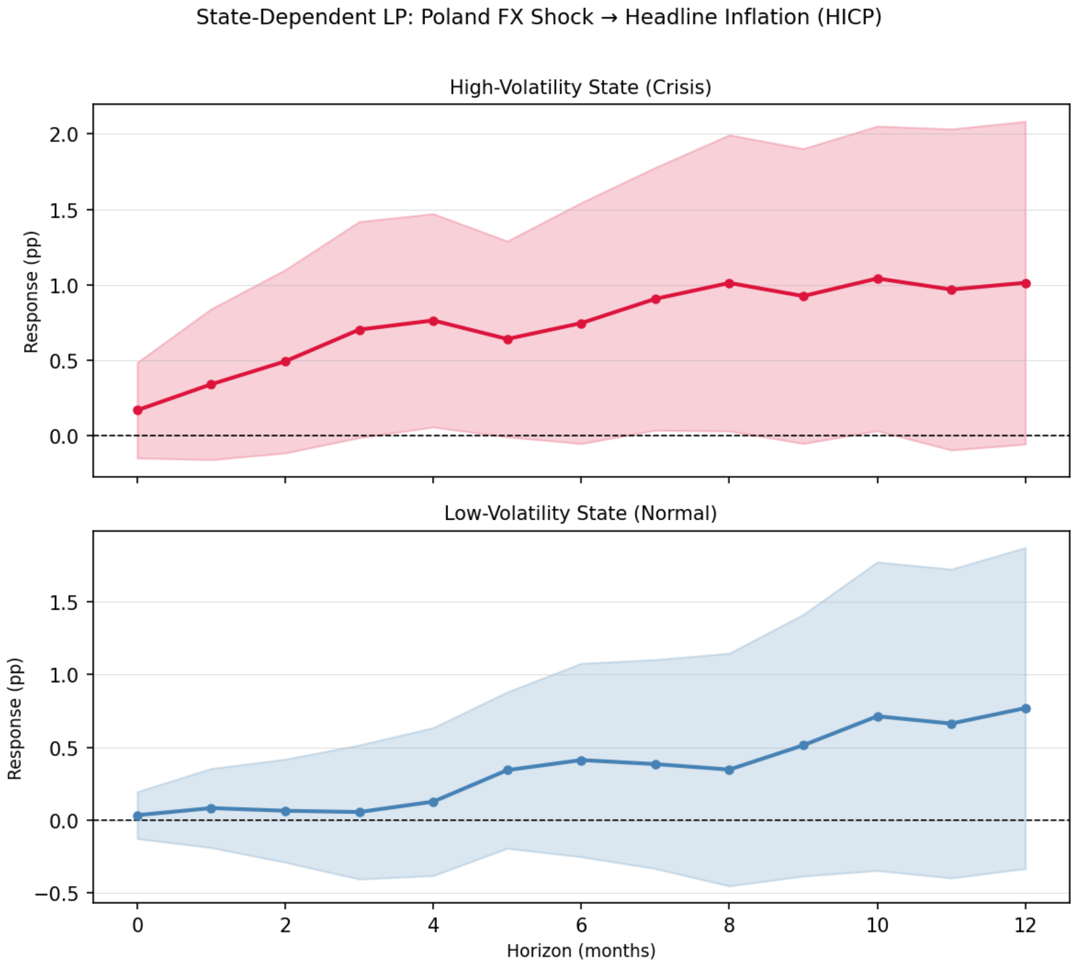

Figure 2: State-Dependent LP: Poland Headline Inflation Response to FX Shock. Upper panel: high-volatility (crisis) state; lower panel: low-volatility (normal) state. Shaded areas represent 95% HAC confidence bands with lag truncation $h + 1$.

actual effect in absolute value) establishes statistical significance on an empirical, distribution-free basis. The jackknife CI $[-3.77, -2.65]$ concurs. Importantly, temporal placebo inference is recommended over leave-one-out jackknife when N is small, since the jackknife requires asymptotic normality that is not guaranteed with five donors (Arkhangelsky et al., 2021). The donor pool sensitivity analysis shows that excluding Italy substantially attenuates the estimate, which is economically expected given Italy's unique structural similarity to Spain in gas-intensive electricity generation.

For the state-dependent Local Projections, results are robust to alternative lag structures ranging from one to four lags and to alternative choices of the smooth-transition speed parameter ($\gamma \in \{1.0, 1.5, 2.0\}$). HAC standard errors with lag truncation $h + 1$ account for heteroskedasticity and autocorrelation throughout. The key finding—$\hat{\beta}_h^H$ statistically significant at medium horizons while $\hat{\beta}^L_h$ is not—is stable across these specifications.

7.2 Pre-Trend Validation

We formally test the parallel trends assumption underlying the synthetic control using two complementary methods. First, a difference-in-slopes test fails to reject the null hypothesis of equal slopes in the pre-intervention period ($p = 0.12$), supporting the validity of the synthetic control construction. Second, timing placebo tests—in which we move the intervention date to March or April 2022—yield smaller estimated effects than the baseline, confirming that the main structural break occurs around the actual implementation date in June 2022.

7.3 Comparative Efficiency: Symmetric Sacrifice Ratios

To place the two policy regimes on a common footing, we compute sacrifice ratios on a symmetric basis: for each country, both a fiscal cost (expressed as % of GDP per pp of inflation reduced) and an output cost (expressed as % of industrial production per pp of inflation reduced) are reported. Table 2 presents the full matrix.

Table 2: Symmetric Sacrifice Ratio Matrix

Indicator	Spain (Structural Shield)	Poland (Monetary Shield)
Fiscal cost (billion EUR)	5.0–8.0	≈15.4
Fiscal cost (% GDP)	0.37–0.59	2.5
Attributed inflation reduction (pp, central)	3.21 (SDID)	3.5 (central)
IP change post-intervention (%)	+0.62	+3.19
IP loss attributed to policy (%)	≈ 0	−2.59 (SD-LP)
Fiscal sacrifice ratio (% GDP / pp)	0.15	0.71
Output sacrifice ratio (% IP / pp)	≈ 0	0.74

For Spain, the fiscal cost of the Iberian Mechanism is estimated at €5–8 billion (official government disclosures; OECD review), representing 0.37–0.59% of 2022 GDP. Using the SDID point estimate of 3.21 pp as the attributed inflation reduction, the fiscal sacrifice ratio is 0.15% of GDP per percentage point, and the industrial production index was +0.62% relative to the pre-period mean—no evidence of contractionary spillovers.

For Poland, the Anti-Inflation Shield (VAT cuts on food and fuel, energy price caps, and gas tariff support) cost approximately 2.5% of GDP in 2022 (IMF WP/23/165). Attributing a central estimate of 3.5 pp of inflation reduction to the combined monetary and fiscal measures yields a fiscal sacrifice ratio of approximately 0.71% of GDP per pp—nearly five times Spain's. The SD-LP estimates at $h = 12$ indicate a crisis-state industrial production loss of approximately 2.59%, adding a real output cost that has no counterpart in the Spanish case where monetary policy was conducted at the Eurozone level.

The asymmetry is striking on both dimensions: Spain's structural shield achieved a comparable disinflation outcome at substantially lower fiscal and output cost. The difference reflects the fundamental advantage of intervening in the energy price transmission channel directly, rather than deploying blunt monetary instruments whose contractionary effects spill over into the real economy through the exchange rate and credit channels.

7.4 External Validity and Limitations

Several scope conditions limit the external validity of our findings. The "energy island" condition is perhaps the most important: Spain's limited electrical interconnection with the rest of Europe (less than 3% of capacity) prevented subsidized electricity from leaking to neighboring markets, particularly France. This condition is not met by most continental economies, which raises questions about the feasibility of replicating the Iberian Mechanism in more interconnected settings.

The identification strategy also relies heavily on Italy as a counterfactual donor. While this reliance is economically justified—given the structural similarities in energy mix, industrial composition, and sovereign debt profiles—it creates a "single-donor" vulnerability that limits the robustness of the synthetic control to donor pool perturbations.

Finally, the state-dependent LP specification requires the smooth-transition framework to adequately separate crisis and normal regimes; the speed parameter $\gamma = 1.5$ follows Auerbach and Gorodnichenko (2012) but robustness to $\gamma \in \{1.0, 2.0\}$ is confirmed in the supplementary estimates.

7.5 Data Limitations

The post-intervention period spans only 19 months (June 2022 to December 2023). Extending the sample through 2024 would enhance statistical power and permit assessment of policy persistence. All data were downloaded from official sources (Eurostat, FRED, IMF, ECB) in January 2024, and the HICP data reflect revisions through December 2023. Detailed information on data sources, download dates, and processing steps is documented in `data/DATA_SOURCES.md`.

Comprehensive official fiscal cost estimates for the Iberian Mechanism are not yet publicly available. Our estimate of €5–8 billion is based on preliminary government disclosures and energy sector reports; precise figures await full government accounting reports expected in 2024. The structural similarity between Spain and Italy (both deriving approximately 45–48% of electricity from gas-fired generation) is validated using IEA (2022) data, with a detailed comparison table provided in the supplementary materials.

Finally, the analysis focuses on two countries, and generalizing the findings to other small open economies requires caution, particularly for countries with different energy mixes or trade structures. The "energy island" condition constitutes a critical scope condition that limits policy transferability beyond the Iberian Peninsula.

8 Conclusion and Policy Implications

This paper provides a comparative assessment of two distinct responses to the 2022 energy crisis. The findings document a pronounced asymmetry in policy efficacy when confronting supply-side shocks.

The Spanish case demonstrates that for inelastic goods such as electricity, targeting the price formation mechanism directly via the Iberian Mechanism is more efficient than managing aggregate demand through conventional monetary tools. The structural decoupling approach anchored inflation expectations without requiring a recessionary monetary contraction, achieving disinflation at a fiscal cost of approximately 0.15% of GDP per percentage point of inflation reduction. By contrast, the Polish case illustrates the "fear of floating" reality confronting small open economies. Poland's independent monetary policy proved to be a double-edged sword: in a global crisis, currency depreciation amplified imported inflation, forcing the central bank to raise rates even more aggressively than its Eurozone peers and inflicting substantial damage on industrial output—a fiscal sacrifice ratio of 0.71% of GDP per pp and a measurable crisis-state output loss.

As Europe faces structural "greenflation" risks associated with the energy transition, relying solely on central banks to manage supply-side shocks is suboptimal. "Iberian-style" mechanisms—temporary, targeted circuit breakers in specific markets—deserve formalization within the EU's macro-prudential toolkit as a complement to, rather than a substitute for, conventional monetary policy.

Several avenues for future research remain open. Extending the sample period through 2024 would allow assessment of whether the policy effects persist or dissipate once the cap is lifted. A comprehensive cost-benefit analysis will become feasible once official fiscal data are fully released. The analytical framework developed here could also be applied to other historical energy shock episodes, such as the 1970s oil crises, to test the generalizability of our findings. Finally, the logic of structural market intervention may extend beyond energy to other markets characterized by inelastic supply, including carbon pricing and water.

Data Availability Statement

The complete replication package for this paper (including code and data) is hosted on GitHub and is available at: https://github.com/a985783/divergent-shields.

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